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Research Paper

A descriptive study on the use of restrictive interventions for potentially or actually violent patients in the emergency department

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ABSTRACT

Background: Restrictive interventions (chemical, physical, or mechanical restraints) to manage patients who are potentially or actually violent in the emergency department (ED) can be harmful and costly. Non-restrictive interventions are advocated; but this must be preceded with an understanding of patient characteristics that influence their use. A study was conducted to describe the use of restrictive interventions and ascribe it with patient characteristics in the ED.
Methods: Records from October 2020 to March 2021 in the occupational violence database were used to analyse patient characteristics and restrictive interventions. Logistic regression was used to establish influencing factors of restrictive interventions adjusting for clinically relevant confounders.
Results: Of the N = 1276 potentially or actually violent patients, 70 % received restrictive interventions. Chemical restraint was common, with 1 in 2 patients receiving either oral medication or intramuscular injection. Probability of restrictive interventions were higher in patients who were intoxicated [(adjusted odds ratio (aOR) 3.48, 95 % confidence interval (CI) 1.675–7.21)], had high triage score (aOR 2.084, 95 % CI 1.094–3.96), and were in the ED involuntarily (aOR 1.494, 95 % CI 1.105–2.020).
Conclusion: The results reveal the need for multifaceted approaches that limit the presentations of, and minimise restrictive interventions among, potentially or actually violent patients.
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Introduction

Occupational violence (OV) is any verbal abuse or physical attack that occurs in the workplace, which could lead to psychological harm or physical injury [1]. It is pervasive in emergency departments (ED) with up to 98 % of healthcare staff having been subjected to verbal OV [1], and about 20 % to physical OV [2]. Due to the harms OV incidents pose [3], early intervention utilising non-restrictive practices is a priority.

Responding to patients who are potentially or actually violent may involve verbal de-escalation [4] and restrictive interventions [5–8]. In this paper, restrictive interventions can be chemical, physical and/or mechanical restraints (see Table 1) [9]. Restrictive interventions should be used as a last resort, however use of restrictive interventions among potentially or actually violent patients range from 24 to 96 % [5–8]. It is not possible to identify the most common type of restraint utilisation, because every healthcare setting varies in terms of incidence of violence, clinical criteria and policy provisions for the use of restraints [10].

The use of restrictive interventions is not without consequence and cost. Chemical restraint is associated with adverse respiratory and cardiac effects ranging in incidence from 6 to 37.5 % [11]. Physical and mechanical restraints in particular can lead to physical and psychological harm in patients [12,13] and staff [14]. Restraints have been described as disempowering and dehumanising [12]. In some patients, restraints invoke lasting fear, anger, frustration, and resentment towards health care staff [15]. For other patients witnessing, restraints were described as unpleasant and confronting [15].

From a staff perspective, restrictive interventions cause a clinical quandary between respect for patients' freedom and autonomy, and maintaining patient and personal safety [16]. While restrictive interventions are seen by staff as necessary for safety and 'last resort' to quell violent behaviour [17,18], many have ascribed their experience as 'horrible' and brought feelings of job dissatisfaction [19]. Due to the involved nature of physically or mechanically restraining a patient, such as requiring multiple people to hold a combative person, it poses hazards to staff. In one report, 25 % of staff injuries occurred during their involvement in restraint procedures [14]. As to

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Table 1
Description of restrictive interventions [4] and their examples.

Chemical restraint	medications (oral, intramuscular or intravenous) are administered with the desired outcome of sedation rather than symptom management. In the study ED, they can include diazepam, lorazepam, olanzapine, or droperidol).
Physical restraint	patient is controlled by other individuals holding parts of the patient's body.
Mechanical restraint	patient's movement is restricted by equipment or use of various forms of strapping (i.e. handcuffs, soft leather restraints, taser)

the cost, excluding the staff sick leave and compensation costs, the financial impact of restrictive interventions is about AU$400,000 (2014 dollar value) per year, per ward from medications (i.e. sedatives) and labour [20].

In the setting of consequences and cost of restrictive interventions, there have been emphasis on patient-centred approaches to reduce its use [21]. However, these approaches may not materialise without the ability to identify patient characteristics. As highlighted above, there is a reasonable volume of literature containing the prevalence, consequences, and cost of restrictive interventions; but patient characteristics that may influence their utilisation is lacking. The importance of understanding patient profile is two-fold. First, it leads to the identification of who are at-risk of receiving restrictive interventions. Second, it could facilitate working towards non-restrictive approaches to managing violence. Therefore, the objectives of this study were to: characterise patients who received restrictive interventions in the ED; identify patient characteristics associated with restrictive intervention use; and compare resource use between those who did and did not have restrictive intervention/s.

Ethics

The study was approved by the Metro South Human Research Ethics Committee (HREC/2021/QMS/75412).

Methods

Study design

The study was an observational study, which was guided by the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement [22].

Setting

The ED is a public, metropolitan, adult tertiary referral hospital in Brisbane, Australia that had over 69,000 presentations in 2021. The ED has a mental health unit and a specialist toxicology unit. Based on in-house security reports, the ED has a disproportionately high volume of OV incidents compared to other clinical areas in the hospital. The ED has a round-the-clock, nurse-led team - Response to Occupational Violence Emergencies (ROVE) - with advanced practice nurses who respond to potential or actual OV incidents with security personnel. The ROVE team received the education and training programme on managing OV that is supported by the health service [23]. They are primarily stationed at triage to maximise early identification of violence risk (i.e. agitation score, verbal aggression, physical violence), physical oversight and timeliness of response. ROVE can respond automatically or when triggered by a healthcare clinician or security for a perceived need (i.e. risk of OV that is beyond their capacity to de-escalate). The ROVE team can verbally de-escalate and initiate restrictive interventions (Table 1) - to patients who are at-risk of perpetrating violence or already violent in the ED. Use of mechanical restraints are decided collaboratively and authorised by the treating doctor. When deemed necessary, a de-escalation room is used by the ROVE team.

Data collection

The study population were identified from the ROVE database between October 2020 and March 2021 (six months). ROVE nurses maintain this database with patient identifiers, demographics, date and time of ED arrival, date and time of response, intervention/s provided (verbal de-escalation only, oral chemical restraint, intramuscular (IM) or intravenous (IV) restraint, physical restraint, mechanical restraint), total time on task, number of security personnel in attendance, and number of clinical staff (nurse/s and/or doctor/s) in attendance. In this database, one entry represents a new response, so there may be more than one entry if a patient whom the ROVE team had to manage had more than once in a single ED presentation. Only the first response was included in the study to avoid confounding the effect of preceding ROVE management.

Using the patient identifier and ED arrival date and time, the ROVE database was linked to the FirstNet database (electronic medical records system in the ED) to include emergency care variables such as mode of arrival, voluntary status (under Chapter 4A of the Public Health Act 2005, Qld) [24], indigenous status, triage category, ED length of stay, and ED disposition. Finally, data about prehospital restrictive interventions (i.e. initiated by police or ambulance officers), mental health history, Sedation Assessment Tool (SAT) score, and intoxication status (could be objective i.e. breathalyser or blood test or subjective i.e. self-reported by patient) were manually extracted from the ROVE clinician's documentation. Data were excluded from the analysis if the ROVE response could not be linked to an ED presentation or did not occur in the ED. Given the round-the-clock rostering of ROVE, the database captures most of our OV risks and incidents.

Five percent of the dataset were randomly selected for accuracy check by two investigators (HT and CJC). We detected a 40 % discrepancy mainly in the ED date and time of arrival, which was a consequence of the linkage formula. All discrepancies in the entire dataset were identified and corrected.

Data analysis

Patients were grouped as 'restrained group' (for brevity) if any chemical, physical, or mechanical restraint was applied; otherwise they were grouped as 'verbal de-escalation only'. Data were analysed in IBM SPSS Statistics V23 (North Sydney, Australia). Statistical tests and p values are reported. A p value of 0.05 is considered statistically significant.

We first described, then compared, patient characteristics – age, older adults (< 65 vs ≥65 years old), mode of arrival, triage category, intoxication status, voluntariness of presentation, evidence of prehospital restraint, and mental ill-health history - between those who were and were not managed with coercive restraints (verbal de-escalation only versus any restraints). All variables (except for age) were categorical variables and were analysed using Chi-square test. Age was analysed using Mann-Whitney U test. Ideally, SAT scores would have been included as a predictor, but due to substantial missing data these were omitted.

Standard logistic regression was conducted to establish associations between patient characteristics and restraint use. Patient variables included in the model were based on the results of the bivariate analysis described in the paragraph above. This method of variable selection ensures that predictor variables have clinical

rather than theoretical influencing value on the outcome [25]. An odds ratio with a value greater than 1 is interpreted as an increase in odds of restraint use. Inversely, an odds ratio less than 1 translates to a decreased probability of restraint use. Odds ratios with corresponding 95 % confidence intervals (CI) that do not include the value of 1 are statistically significant.

Finally, we explored resource implications such as number of security and clinical staff required to manage the patient, length of ROVE management, ED disposition, and ED length of stay. Continuous outcomes (number of staff, length of ROVE management, and ED length of stay) were analysed using Mann-Whitney U test; otherwise, categorical outcome (ED disposition) was analysed using Chi-square test.

Results

Population characteristics

There were N = 1276 presentations seen by ROVE included in the analysis. This translates to 3.68 per 100 ED presentations being seen by the ROVE team over the study period. Overall, 15.44 % (n = 197) received any form of pre-hospital restrictive intervention.

Overall, 70 % (n = 893) of patients who were potentially or actually violent in the ED received some type of restraint (see Fig. 1). About 1 in 3 received more than one form of restraint. Oral chemical restraint was given more commonly than other restraint type (n = 493, 55.2 %), followed by IM or IV chemical restraint (n = 423, 47.4 %), physical restraint (n = 282, 31.9 %), and mechanical restraint (n = 24, 2.7 %).

Comparing the characteristics of those who received verbal de-escalation only (N = 383) and those were restrained (N = 893), the restrained group were significantly more likely to: be male, present via ambulance and/or police (p = 0.001), have a higher triage score (p < 0.001), present to the ED involuntarily (p < 0.001), be intoxicated (p < 0.001), and have a history of substance use disorder (p = 0.008) (See Table 2).

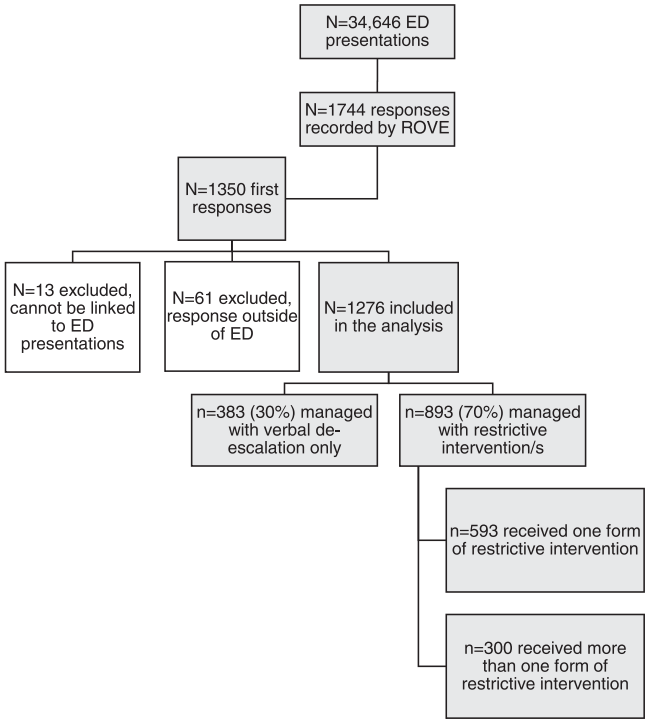

Fig. 1. Study flow chart depicting the size of the study population between October 2020 and March 2021 that were included in the analysis. Legend: ED, emergency department; ROVE, Response to Occupational Violence Emergencies; restrictive interventions include chemical, physical, or mechanical restraints.

Table 2
Comparison of characteristics of the study population.

	Verbal de-escalation group (N = 383)		Restrained group (N = 893)			
	n	%	n	%	χ^2	*p*
Age (median, range)	35 (16–89)		34 (15–95)		–	0.273
Older adults (≥65 years)	18	4.7 %	40	4.5 %	0.30	0.862
Gender						
Female	127	33.2 %	328	36.7 %	1.49	0.222
Male	256	66.8 %	565	63.3 %		
Mode of arrival						
Walk-in	68	17.8 %	109	12.2 %*	16.166	0.001
Ambulance	231	60.3 %	500	56.0 %		
Police	53	13.8 %	189	21.2 %*		
Ambulance and Police	31	8.1 %	95	10.6 %		
Australian Triage Scale						
1	22	5.7 %	91	10.2 %*	23.321	< 0.001
2	81	21.1 %	245	27.4 %*		
3	229	59.8 %	495	55.4 %		
4 & 5	51	13.3 %	62	6.9 %*		
Presentation type						
Voluntary	213	55.6 %	372	41.7 %	21.028	< 0.001
Involuntary	170	44.4 %	521	58.3 %		
Intoxication status (can be self-reported by patient, breathalyser or blood test result)						
None	242	64 %	386	43.3 %*	49.509	< 0.001
Alcohol	82	21.7 %	254	28.5 %*		
Drugs	45	11.9 %	195	21.9 %*		
Alcohol and drugs	9	2.4 %	56	6.3 %*		
Pre-hospital restrictive intervention (i.e. initiated by police or ambulance officers)						
Received	47	14.9 %	150	19.1 %	2.750	0.097
History						
Aggression	161	42.9 %	344	39.2 %	1.501	0.220
Anxiety	62	16.2	165	18.5 %	0.960	0.327
Neurodevelopmental disorder	22	5.7 %	54	6 %	0.044	0.834
Neurocognitive disorder	33	8.6	55	6.2	2.52	0.112
Eating disorder	9	2.3 %	13	1.5 %	1.265	0.261
Mood disorder	118	30.8 %	289	32.4	0.298	0.585
Personality disorder	50	13.1 %	99	11.1 %	1.007	0.316
Schizophrenia and psychotic disorder	65	17 %	181	20.3 %	1.873	0.171
Substance use disorder	120	31.3 %	350	39.2 %	7.121	0.008
Trauma-related disorder	38	9.9 %	105	11.8 %	0.908	0.341

*, statistically significant.

Influencing factors for any restraint use

Intoxication status, triage score, mode of arrival, patient's voluntariness and history of substance use disorder were included in the regression analysis. Three factors remained as the independent factors for any restraint use: patients' intoxication status, triage score, and voluntariness. Alcohol and/or drug intoxication increased the odds of restraint by up to 3.48 times (95 % CI 1.675 – 7.21, p = 0.001). Compared to those with a triage score of 4 or 5, a triage score of 1 increased the odds of restraint 2.084 times (95 % CI 1.094 –

Table 3
Results of logistic regression showing association between patient and presentation characteristics and restraint use.

Predictor	adjusted odds ratio	95 % CI	*p*
Intoxication status (can be self-reported by patient, breathalyser or blood test result)			
None	reference		
Alcohol	1.654	1.214 – 2.252	0.001
Drug	2.448	1.689 – 3.549	< 0.001
Alcohol and drug	3.475	1.675 – 7.211	0.001
Triage score			
4&5	reference		
1	2.084	1.094 – 3.967	0.025
2	1.733	1.074 – 2.796	0.024
3	1.268	0.822 – 1.955	0.283
More of arrival			
Walk-in	reference		
Ambulance	0.970	0.659 – 1.428	0.876
Police	1.296	0.776 – 2.166	0.321
Ambulance and police	0.946	0.822 – 1.955	0.858
Presentation type			
Voluntary	Reference		
Involuntary (Chapter 4A of the Public Health Act 2005 (Qld))	1.494	1.105 – 2.020	0.009
Substance use disorder			
No	Reference		
Yes	1.230	0.940 – 1.609	0.132

3.96, p = 0.025). Being involuntary in the ED increased the odds of restraint by 1.494 times (95 % CI 1.105 – 2.020, p = 0.009) (Table 3).

Outcomes of restraint use

The median number of staff required and the time of OV management were similar between verbal de-escalation group and restrained group (Table 4). However, the length of ED stay in the restrained group was 36.6 min longer (p < 0.001). The restrained group were more likely to be admitted to the short stay unit, while the verbal de-escalation group were more likely to discharge themselves against medical advice (p < 0.001).

Discussion

Two-thirds of patients who were potentially or actually violent received restrictive interventions, commonly with medications taken orally. Key patient characteristics associated with restrictive interventions were intoxication, a triage score of 1 or 2, and involuntary attendance to ED. In a previous study [8], only the severity of agitation was known to influence the use of restrictive interventions. Consolidating these findings, patients who have severe agitation, are intoxicated, have high triage score, and are involuntary are at-risk of receiving restrictive interventions in the ED. This understanding draws two important discussion points relating to prevention of ED presentations by, and management of, intoxicated persons when they pose a violence risk or are violent. The discussion follows in the order of the patient journey from the community to the ED to highlight a shared responsibility of care between public health, pre-hospital care, and ED.

The increase in ED presentations of intoxicated persons has been previously highlighted [26,27], which indirectly signals increasing uncontrolled alcohol consumption and drug use in the community. Together with our findings, we believe that there is value for public health interventions that could reduce presentations of intoxicated persons to the ED such as alcohol control policies (i.e. advertising and taxation) that could potentially limit alcohol consumption [28], and other harm minimisation interventions [29].

Acknowledging that alcohol and drug use is unlikely to be eliminated at the public health level, pre-hospital and emergency care of this cohort remain relevant. Further, from a legal standpoint if a person's behaviour is deemed a risk to themselves (e.g. from severe intoxication) and community, ambulance officers can lawfully refer the person to the ED or their home (whichever is deemed appropriate) [24]. Sobering centres could be explored as a plausible alternative to ED presentations;[30] but in their absence, care of intoxicated persons will remain the responsibility of ambulance officers and emergency staff. This situation draws the need for pre-hospital de-escalation to occur first.

De-escalating intoxicated persons can be challenging because their perception of environmental threat are exaggerated and so does their response to the threat that could include unpredictability and violence [31]. De-escalation principles are thoroughly described elsewhere [4,32], but for completeness they can include these principles: (1) Maintain a safe space (at least 2 arms length from the patient); (2) Avoid provocation; (3) Establish verbal contact; (4)

Table 4
Differences in resource use between verbal de-escalation and restraint group.

	Verbal de-escalation only (N = 383)		Restrained group (N = 893)		z-score	*p*
Number of clinical staff required	2 (1–6)		2 (0–10)		0.44	0.66
Number of security staff required	2 (0–6)		2 (0–5)		1.356	0.175
Length of ROVE management	20 (0 – 120) minutes		20 (0–195) minutes		0.991	0.322
ED length of stay	3.27 (0.05–28.72) hours		3.88 (0.07–35.25) hours		3.988	< 0.001
Disposition following ED care						
	n	%	n	%	χ^2	*p*
Home	129	33.7 %	204	22.8 %*	94.032	< 0.001
Short Stay Unit	74	19.3 %	378	42.3 %*		
Ward	45	11.7 %	86	9.6 %		
Mental Health Unit	87	22.7 %	197	22.1 %		
Left against medical advice	43	11.2 %	23	2.6 %*		
Other health facility	5	1.3 %	5	0.6 %		

ROVE, Response to Occupational Violence Emergencies; ED, emergency department; *, statistically significant.

Keep communication concise and simple; (5) Acknowledge patient's wants and feelings; (6) Listen closely to the patient; (7) Do not argue, agree to disagree; (8) Set boundaries; (9) Offer choices including medications; and (10) Counsel and de-brief [4].

It is apparent that de-escalation requires skill to be successful; and so the need for education and training of ambulance officers [33] and ED staff [34,35]. Despite this understanding, there have been suggestions that there are de-escalation skill deficits among ambulance officers [36,37] and emergency staff [34,38] that might be aggravating rather than de-escalating patients. There is a possibility that when patients arrive in the ED they could be much harder to verbally de-escalate and the likelihood of its success is smaller; hence the need for restrictive interventions in the ED. Nevertheless, this highlights a shared responsibility between ambulance officers and ED staff to be prepared and possess the skill to de-escalate patients as soon as possible [33–35].

While there is merit for education and training to upskill ambulance officers and ED staff, they are not the panacea because they are ineffective in reducing restrictive interventions and violent episodes in the absence of other strategies [39]. Other strategies could include violence risk assessment tools, availability of non-pharmacological comfort measures, security presence, behavioural management protocols including chemical restraints, allocative strategy (right location, right staff), behavioural management teams, and de-escalation rooms [35]. There is evidence that when education and training are bundled with other strategies they could reduce restrictive interventions [40] and violence in EDs [41–43].

The impact of restraint on staff resources, given the presence of the ROVE clinicians, was minimal. Those who were restrained did not significantly require more time and staff to be managed. One could assume that restrained patients require more staff, but our data show otherwise - the number of personnel is unchanged. This may be due to the ROVE model of care that usually has a standard number of personnel who responds to OV and the collaborative nature of the OV response involving the primary nurse and medical staff [44]. This staffing requirement is comparable with the existing literature [45].

The impacts of restraint on disposition were evident, as this group stayed in the ED 36.6 min (median) longer and are frequently transferred to the observation unit within the ED than those who received verbal de-escalation only. Factoring their intoxication and restraint status, they might stay longer until they are sober and are deemed safe for discharge. The added 36.6 min in length of stay is clinically important, which is likely to impact on ED efficiency and service delivery [46].

Limitations

Study design

As with any retrospective observational studies, there is a risk of selection bias that could skew the results towards or away from the null hypothesis. For example, the study was conducted in an ED that offer toxicology and mental health services, this could explain the prevalence of intoxication in the study population. Being a single site study with the ROVE model of care, the applicability of our findings in other EDs needs to be interpreted with caution.

Data integrity

Retrospective studies are subject to data inaccuracies; and this was mitigated by randomly checking 5 % of the dataset for accuracy and entry errors. We were unable to account for agitation severity using SAT scores due to missing data. The implication of this is that an objective, clinical indication for restraints remains unknown. Since this study, we have made modifications to our work process so that we are now able to document agitation scores promptly.

Other considerations and research implications

Patient characteristics were the main explanatory variables explored in this study. It was beyond the ambit of this study to account for clinician-related variables (i.e. years of experience, training), but this could be argued as a limitation. Future studies could account for this important predictor that may direct the need for education and training of staff who manage patients who could be, or are violent in the ED. Another argument for limitation is the uncertainty around consent for chemical restraint. Chemical restraint can be given to consenting and non-consenting patients [47], or they could be given it as part of their regular medications. We were unable to make this distinction, which would have provided clarity due data limitations, for which future studies could account.

Conclusion

Restrictive interventions, particularly chemical restraint, are commonly used in the ED to manage patients who are potentially or actually violent. Their use is influenced by a patient's intoxication status, triage score, and voluntariness. Patients who had restrictive interventions stay longer in the ED, which is likely to impact ED efficiency. Awareness of these characteristics and time imposition prompt the need for multifaceted strategies that could include public health interventions to limit presentations of intoxicated persons, upskilling of ambulance officers and emergency staff in verbal de-escalation, and bundled management approach in the care of patients who are potentially or actually violent in the ED.

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Conflict of interest

The authors have no conflict of interest to declare.

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